

PRISCILLA SARFOA ANIM

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Priscilla Anim



Priscilla S. Anim

PROFESSIONAL SUMMARY

Statistician and data scientist with extensive experience in statistical modeling, causal inference, and applied machine learning. Proficient in advanced statistical and quantitative methods, with strong computational skills in Python, R, Stata, SQL, and LaTeX. Demonstrated expertise in large-scale data analysis, data cleaning, exploratory analysis, and communicating complex quantitative findings to technical and non-technical audiences. Experienced in collaborative research, survey-based data collection, and applying rigorous quantitative methods to investigate research questions and inform evidence-based decision-making. Seeking advanced methodological training in statistical and quantitative approaches to educational research to contribute to research on learning, talent, and giftedness, and to pursue doctoral studies.

TECHNICAL SKILLS

Programming & Software: Python (Pandas, NumPy, scikit-learn), R, SQL, Stata, SPSS, LaTeX, Git/GitHub

Statistical & Econometric Methods: Difference-in-Differences, Two-Way Fixed Effects, Bayesian inference and causal impact analysis, time-series analysis (ARIMA), regression analysis, statistical inference, experimental design, A/B testing, multivariate distributions, econometrics

Machine Learning: Tree-based models, ensemble methods (XGBoost, Random Forest, LightGBM, CatBoost), PCA, SHAP/LIME interpretability, predictive modeling, feature engineering

Data Visualization: Power BI, Matplotlib, Seaborn, Plotly, dashboards

Research Skills: Literature review, academic writing, qualitative coding, survey design, data collection and validation, analytical memos

EDUCATION

Grand Valley State University, Allendale, MI

Expected May 2027

M.S. in Data Science & Analytics, GPA: 3.88/4.0

- **Relevant Coursework:** Machine Learning, Data Engineering, Statistical Modeling, Generative AI, Statistical Computing and Graphics with R, Multivariate Data Analysis, Knowledge Discovery and Data Mining

University of Ghana, Accra, Ghana

Sep 2019 – Sep 2023

B.A. in Statistics and Economics (Honors)

- **Undergraduate Project:** *Intervention Analysis of Electronic Payment Systems in Ghana – COVID-19 and Electronic-Levy*
- **Advisor:** Dr. Enoch Sakyi-Yeboah

RESEARCH EXPERIENCE

Graduate Research Assistant

May 2026 – Present

Grand Valley State University, Finance Department

Supervisor: Professor Vijay Gondhalekar

- **Academic Plagiarism in Journal Publication (retraction analysis):** Built a Python/Pandas pipeline to clean and restructure Retraction Watch data for U.S.-based retractions, 4,197 records filtered from an initial 70,131, splitting and interleaving multi-author and multi-institution fields and one-hot encoding subject, retraction nature, and reason-for-retraction categories.
- Applied the `gender_guesser` library to classify author gender from first names across all author fields, producing an overall distribution and exporting unresolved names with institutional affiliations for follow-up review.
- Constructed a state-level “Right-to-Work” (RTW) legal-status dummy variable for each author, linking institution state and original-publication year against historical RTW enactment dates.
- Conducted institutional and geographic frequency analysis across 51 states and 6,741 unique institutions and analyzed time-to-retraction patterns, producing static and interactive choropleth visualizations.
- **Pro-ESG and Anti-ESG State Legislative Language Analysis:** Conducted parallel n-gram clustering analyses (unigrams, bigrams, trigrams) on state-level legislative keyword matrices, applying PCA and K-Means clustering with silhouette-score-based optimal cluster selection.
- Used cosine similarity and K-Nearest Neighbors to assess state-level linguistic proximity and conducted cross-n-gram consistency analysis, producing heatmaps, PCA scatter plots, and network-based visualizations.
- **Tools and languages:** Python (Pandas, scikit-learn, Matplotlib, Seaborn, Plotly)

Graduate Research Assistant

Grand Valley State University, Economics Department

Aug 2025 – May 2026

Supervisor: Professor Tiffanie Perrault

- Designed and implemented a Difference-in-Differences (DiD) and Two-Way Fixed Effects (TWFE) framework to estimate the causal effect of staggered state-level cannabis legalization on cannabis prices, merging the THMQ Leafly retail pricing dataset with a state policy adoption timeline.
- Applied and evaluated the Borusyak-Jaravel-Spiess (BJS) imputation estimator to address known bias in traditional TWFE models under staggered treatment timing; complemented the analysis with standard TWFE regressions as a robustness check.
- Conducted data cleaning, cross-source merging, and aggregation of transaction-level pricing data to the state-month level in Stata; produced comparative price-trend visualizations in Stata and Python.
- Produced formatted regression comparison tables (price levels and log-price specifications) using Stata's `esttab` for reporting and reproducibility.
- Collaborated with faculty supervisor Dr. Tiffanie Perrault (Grand Valley State University) and Dr. Rodrigo Aranda and graduate research assistant Gashaw Mekonen (Western Michigan University).
- **Tools and languages:** Stata, Python (Matplotlib), LaTeX

Data and Inventory Analyst

Kharis Management Consult Ltd.

Sep 2024 – Aug 2025

- Conducted statistical analysis of inventory and stock-level data to forecast demand, identify purchasing patterns, and support data-driven supply chain decisions.
- Developed and maintained inventory databases and analytical workflows to organize, monitor, and track stock data using Power BI and Excel.
- Built interactive Power BI dashboards and reports to visualize inventory trends, stock levels, discrepancies, and operational performance for management decision-making.
- Collaborated with procurement teams to analyze inventory requirements and support timely replenishment of materials, minimizing potential stock shortages.
- Analyzed inventory data and prepared periodic analytical reports, translating quantitative findings into actionable insights for management.
- Monitored and analyzed inventory discrepancies to identify variations between recorded and actual stock levels and support improvements in inventory accuracy and stock management.

Research and Teaching Assistant

University of Ghana, Department of Statistics

Oct 2023 – Aug 2024

- Tutored students in Advanced Regression Analysis, Probability Distributions, Sampling Survey, and Survey Methods and Organization.
- Offered one-on-one mentoring, contributing to improved student performance.
- Guided undergraduates through research projects and final theses, providing feedback on methodology and data analysis in SPSS and R.
- Prepared and graded assignments, exams, and class exercises to maintain consistency and fairness.

Field Enumerator

Ghana Baseline Evaluation Survey, University of Ghana

Aug 2023 – Oct 2023

Accra, Ghana

- Managed field data collection from 200+ households for a community development impact evaluation funded by Spark Micro-grant, USA.
- Designed and implemented rigorous data validation protocols, achieving 95% data accuracy and a 100% survey completion rate.
- Coordinated a team of 10 enumerators, ensuring quality control and adherence to research protocols throughout data collection.
- Performed statistical analysis on large-scale survey data using Stata and R to inform evidence-based policy recommendations.
- Developed comprehensive data collection frameworks and training materials for field staff.

CURRENT RESEARCH PROJECTS

Global Primary Education System Efficiency: Enrollment–Completion Gaps, 1996–2024

Analysis 2026

R, Statistical

- Analyzed global primary education enrollment and completion data across countries and income groups to examine disparities in educational system efficiency and progress over time.
- Found substantial enrollment–completion disparities between low-income and high-income countries, with mean gaps of 47.87 and 1.77 percentage points, respectively; ANOVA confirmed significant differences across income groups.
- Applied regression and convergence analysis to assess trends, finding that gaps declined by approximately 0.46 percentage points annually.

The Socioeconomic Landscape of Gun Violence in America: A State-Level Analysis of Poverty as a Predictor of Firearm Violence, 2014–2023 *R, Statistical Inference 2026*

- Integrated 286,420 incident-level records from the Gun Violence Archive with poverty, income, and population estimates from the U.S. Census Bureau’s American Community Survey, aggregating to the state-year level (N = 500).
- Using a nonparametric permutation test (10,000 iterations), found that high-poverty states averaged 26.28 gun violence incidents per 100,000 residents versus 13.60 in low-poverty states.

Predictive Modeling and Statistical Inference of Adolescent Risk Behavior: Evidence from Three Decades of CDC Surveillance Data *Python, Ensemble Machine Learning, SHAP 2026*

- Collaborative project using the CDC Youth Risk Behavior Surveillance System (1991–2017, 1.17M+ records); independently managed data cleaning, preprocessing, exploratory analysis, and project scoping.
- Applied Random Forest, Gradient Boosting, XGBoost, and CatBoost with SHAP-based interpretability, identifying statistically significant sex-based disparities in risk-behavior prevalence and alcohol use as the most consistently prevalent risk category across survey years.

EDUCATIONAL SERVICE

- **Peter Cook Leadership Academy, Grand Valley State University, 2025–2026** — Selected as a graduate participant in this leadership development program sponsored by the Peter Cook Family; curriculum covered public speaking, self-awareness, collaboration, and confidence-building.
- **Resident Assistant, 2026–Present** — Serve on-call duty rotation and foster a safe, inclusive residential community; respond to health and safety concerns and connect students to campus support resources.

CONFERENCES ATTENDED

- Women in Statistics and Data Science Conference, 2026
- Symposium for Data Science and Statistics, 2026
- Joint Statistical Meetings, 2026
- Workshop — Curry College (Institute for Introductory Statistics Instructors), 2026
- ENAR Spring Meeting, 2026

PROFESSIONAL ASSOCIATIONS

- American Statistical Association (ASA)
- Association of African Quantitative Scientists (AAQS)
- Rewriting The Code (RTC)
- Women in Machine Learning and Data Science (WiMLDS)
- Women Who Code (WWCode)

HONORS, AWARDS, AND SCHOLARSHIPS

- Peter Cook Leadership Academy — selected graduate participant, Grand Valley State University, 2026
- Diversity Mentorship Program Travel Award, 2026
- Curry College Workshop for Academic Statisticians Travel Award, 2026
- ENAR Spring Meeting Travel Award, 2026

CERTIFICATES

- Data Analyst Certificate — Blossom Academy
- Associate Data Engineer — DataCamp
- Data Analyst Associate — DataCamp
- Business Analytics & Data Visualization — iiAfrica